COMPARATIVE ANALYSIS OF WINDOW-BASED FIR FILTERS FOR NOISE REDUCTION

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1.Abstract

This report presents the design and comparative analysis of Finite Impulse Response (FIR) filters using different window techniques for noise reduction in digital signals. FIR filters were designed using five types of windows: Rectangular, Hamming, Hanning, Blackman, and Kaiser. A noisy signal was generated by adding Gaussian white noise to a 50 Hz sinusoidal signal, and each filter was applied to evaluate its noise reduction capability.

Both qualitative and quantitative analyses were performed. The results show that the Kaiser window (with $\beta=14$) provided the highest noise suppression, achieving an SNR improvement of 6.61 dB and the lowest RMSE of 0.2287. In contrast, the Rectangular window was the least effective, yielding a 6.28 dB improvement. These findings confirm that window selection is critical for optimizing the trade-off between transition width and stopband attenuation, as established in the literature [1][3].

2.Introduction

In real-world signal processing applications, signals are often affected by unwanted noise during acquisition, transmission, or processing. This noise degrades signal quality and makes analysis difficult. Therefore, noise reduction is an essential step in digital signal processing (DSP).

Digital filters are widely used to remove noise. Among them, FIR filters are preferred because they are stable and have a linear phase response, which preserves the shape of the signal without introducing phase distortion [3].

The window method is a simple and effective technique for designing FIR filters. In this approach, an ideal impulse response is truncated using a window function to obtain a practical filter. However, the performance of the filter depends heavily on the type of window used, as different windows offer different trade-offs between transition bandwidth and side lobe attenuation.

This project aims to study and compare different window functions—Rectangular, Hamming, Hanning, Blackman, and Kaiser—and analyze their impact on noise reduction performance when used in FIR filter design [1].

Literature Review

Before starting the implementation, some research papers and textbooks were referred to understand FIR filters and window methods. From the literature, the following points were noted:

- Rectangular window has sharp transition but high ripple
- Hamming and Hanning windows reduce ripple
- Blackman window provides better stopband attenuation
- Kaiser window is adjustable using a beta parameter

These studies, primarily based on Harris [1] and Oppenheim & Schafer [3], form the theoretical basis for this project.

3. Methodology

3.1 Signal Generation

A 50 Hz sinusoidal signal of unit amplitude was generated with fs = 1000 Hz over 1 second (1000 samples). Gaussian white noise ($\mu=0$, $\sigma=0.5$) was added using NumPy's ‘random.randn’ function with seed 42 for reproducibility.

3.1.1 FIR Filter Equation

An FIR filter works by taking a weighted sum of the current and past input samples. The output y[n] is given by:

$$y[n] = \Sigma\, h[k] \cdot x[n-k] \text{ for } k = 0 \text{ to } N-1$$

where:

- x[n] is the input signal

- h[k] are the filter coefficients (determined by the window function)

- N is the filter order

In this project, the filter coefficients were generated using the window method, and the convolution was implemented using SciPy's ‘lfilter’ function.

3.1.2 Group Delay Compensation

FIR filters with linear phase introduce a constant group delay of (N-1)/2 samples. For a filter order of 51, the delay is 25 samples. To calculate SNR and RMSE correctly, the filtered output was shifted backward by 25 samples using ‘np.roll()’ before comparing with the original signal.

3.2 Filter Design Parameters

The following parameters were used for filter design:

Parameter	Value
Sampling Frequency	1000Hz
Filter Order	51
Cutoff Frequency	100Hz
Kaiser Beta(β)	14.0

3.3 FIR Filter Design

FIR filters were designed using the window method via the SciPy library's “firwin” function. For each window type, filter coefficients were generated using the appropriate window function:

- Rectangular: window = "boxcar"
- Hamming: window = "hamming"
- Hanning: window = "hann"
- Blackman: window = "blackman"
- Kaiser: window = ('kaiser', 14.0)

3.4 Processing Steps

The following steps were carried out:

1. Clean sinusoidal signal was generated
2. Gaussian noise was added to create a noisy signal

3. FIR filters were designed for each window type
4. Each filter was applied to the noisy signal using linear convolution, and the output was shifted by N/2 samples to compensate for the filter's group delay
5. Performance was evaluated using visual inspection of plots and quantitative metrics including SNR improvement and RMSE

4. Window Functions Analysis

Each window function exhibits distinct characteristics that influence filter performance:

Rectangular Window: This window provides the sharpest transition width but suffers from high side lobe levels (approximately -13 dB) [1]. As a result, it exhibits significant Gibbs phenomenon oscillations and poor noise suppression. In this study, it gave the lowest SNR improvement of 6.28dB.

Hamming Window: With side lobe attenuation of approximately -41 dB [1], this window offers a good balance between main lobe width and side lobe suppression. It produced moderately smooth outputs with an SNR improvement of 6.43dB.

Hanning Window: With side lobe attenuation of approximately -31 dB [1], Hanning provides comparable performance to Hamming, giving nearly identical SNR improvement of 6.43dB.

Blackman Window: Offering side lobe attenuation of -58 dB [1], Blackman significantly reduces noise at the cost of a wider transition band. It achieved the second best SNR improvement of 6.52dB.

Kaiser Window: This window features a tunable β parameter that controls the trade-off between main lobe width and side lobe attenuation [1][3]. With β = 14.0 used in this implementation, it achieves side lobe attenuation of approximately -78 dB, resulting in the best noise suppression among all tested windows with an SNR improvement of 6.61dB.

5.Algorithm

Performance Comparison of Window-Based FIR Filters

Input: Clean signal s[n], noise sequence n[n], filter order N, cutoff frequency fc, sampling frequency fs

Output: Filtered outputs y_w[n] for each window w, SNR improvement metrics

1. Generate noisy signal x[n] = s[n] + n[n]
2. For each window function w in {Rectangular, Hamming, Hanning, Blackman, Kaiser}:
   a. Design filter coefficients h_w[k] using window method

b. Apply filter: $y_w[n] = \Sigma\ h_w[k] \cdot x[n-k]$

c. Compute SNR_before = SNR(x[n], s[n])

d. Compute SNR_after = SNR(y_w[n], s[n])

e. Compute RMSE = $\text{sqrt}\ (1/N\ \Sigma\ (s[n] - y_w[n])^2)$

f. Store results

3. Plot all outputs y_w[n] for visual comparison

4. Generate comparison table with SNR improvement values

6. Implementation Details

The implementation was carried out using Python with the following libraries:

Library	Purpose
NumPy	Signal generation and numerical operations
SciPy	FIR filter design using 'firwin' function
Matplotlib	Plotting and visualization of results

The filters were implemented using the firwin function from SciPy's signal processing module [2]. For standard windows, the window name was passed as a string. For the Kaiser window, a tuple containing the window name and beta parameter was used.

Filtering was performed using the lfilter function, which implements linear convolution. Performance metrics including SNR and RMSE were calculated using custom functions.

7.Results and Analysis

Figure 1: Original vs Noisy Signal Before Filtering

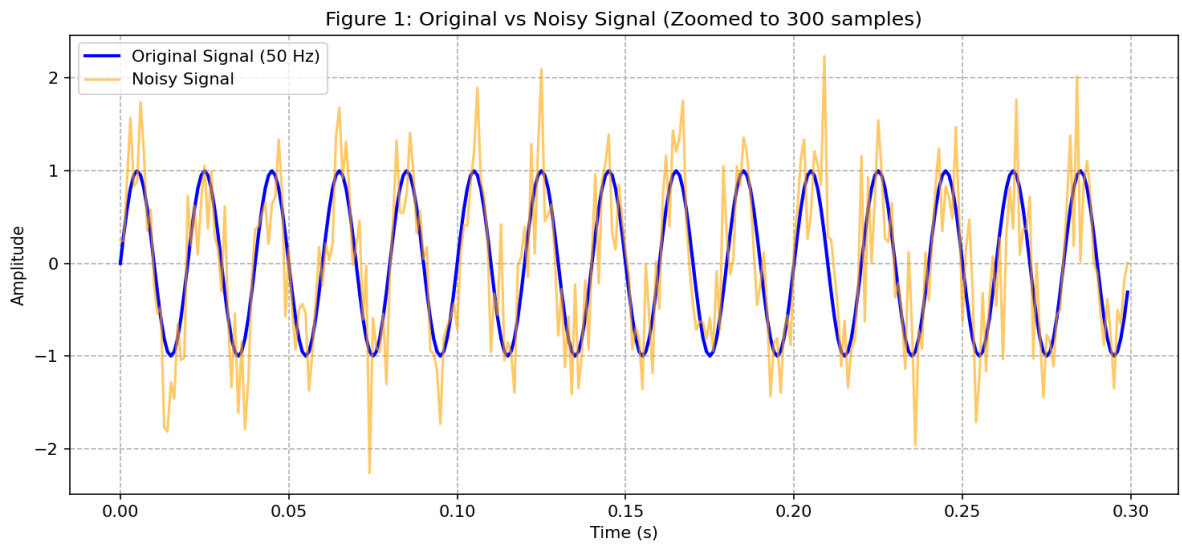


Figure1 shows the original 50 Hz sine wave and the same signal after adding Gaussian noise. The random fluctuations are clearly visible, demonstrating the degradation that occurs in real-world signal acquisition. The input SNR was measured at 3.20dB.

Figure 2: Individual Filter Outputs for Each Window

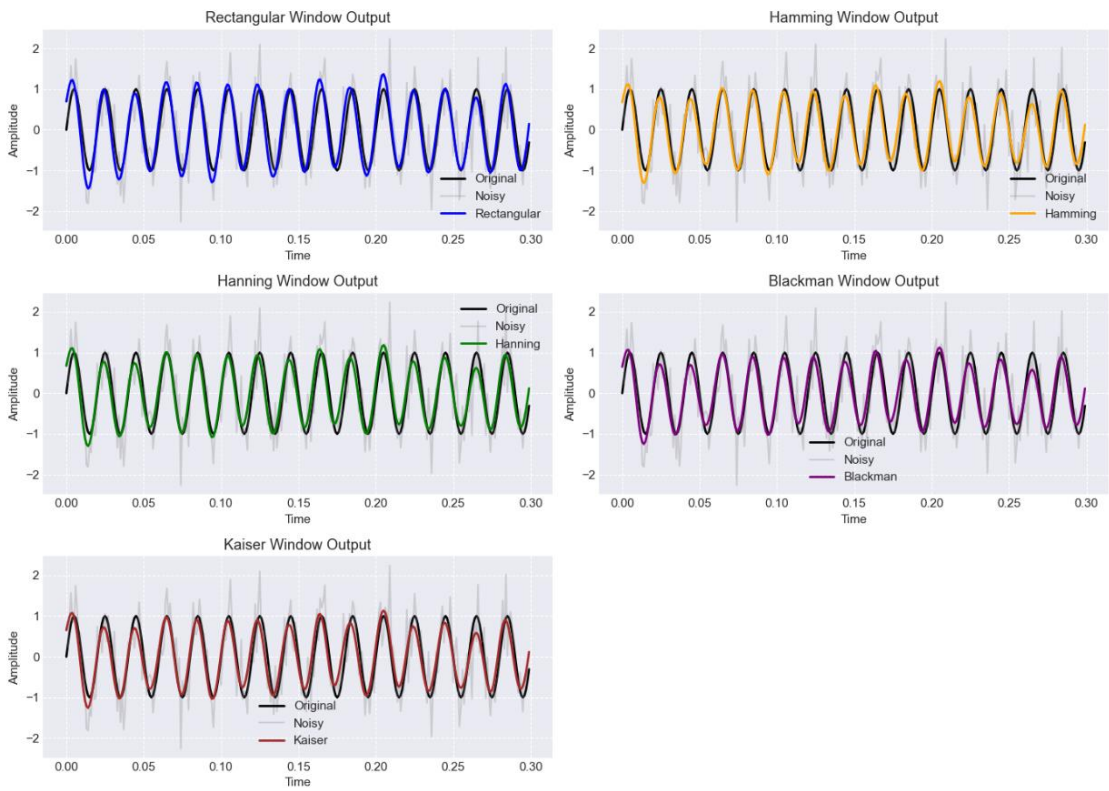


Figure 2 shows each filter's output separately. The Rectangular window output still contains visible noise and oscillations. As we move from Hamming to Hanning to Blackman, the outputs become progressively smoother. The Kaiser window (β=14) produces the cleanest output, most closely matching the original sine wave.

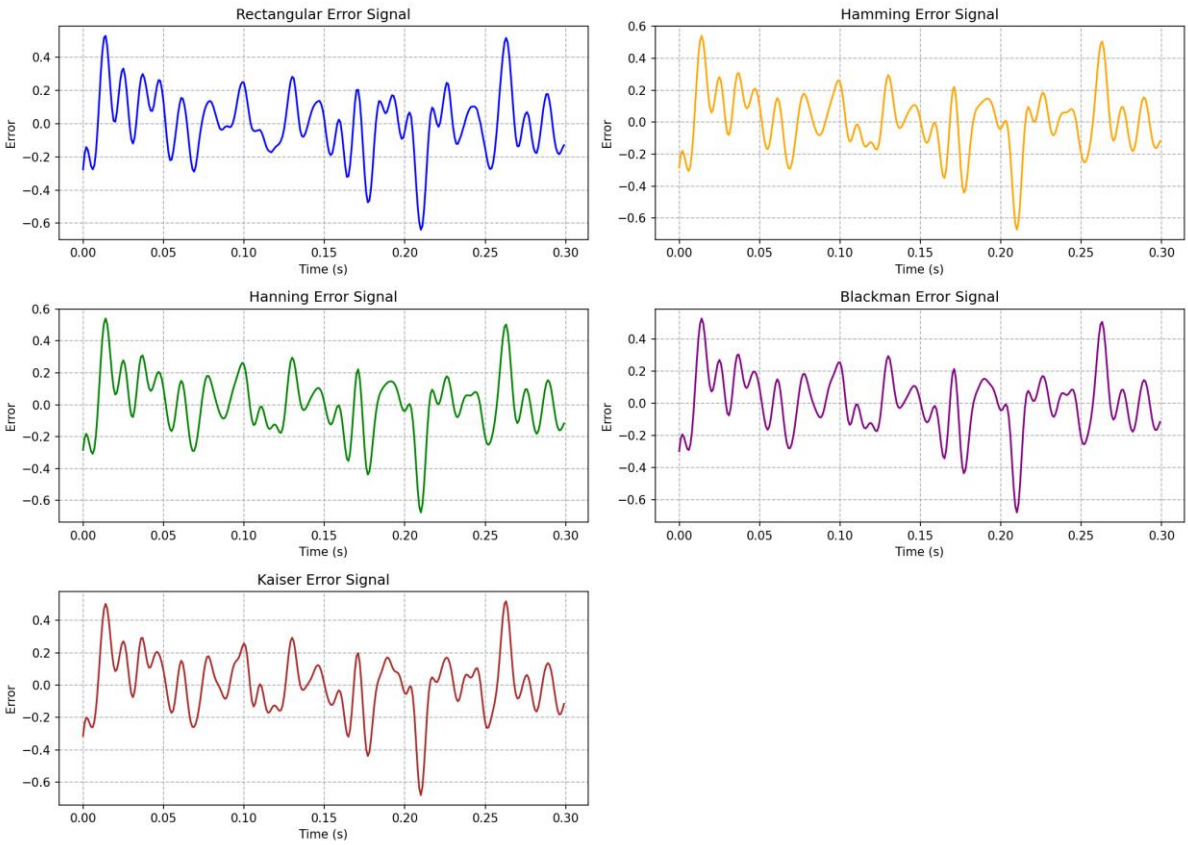
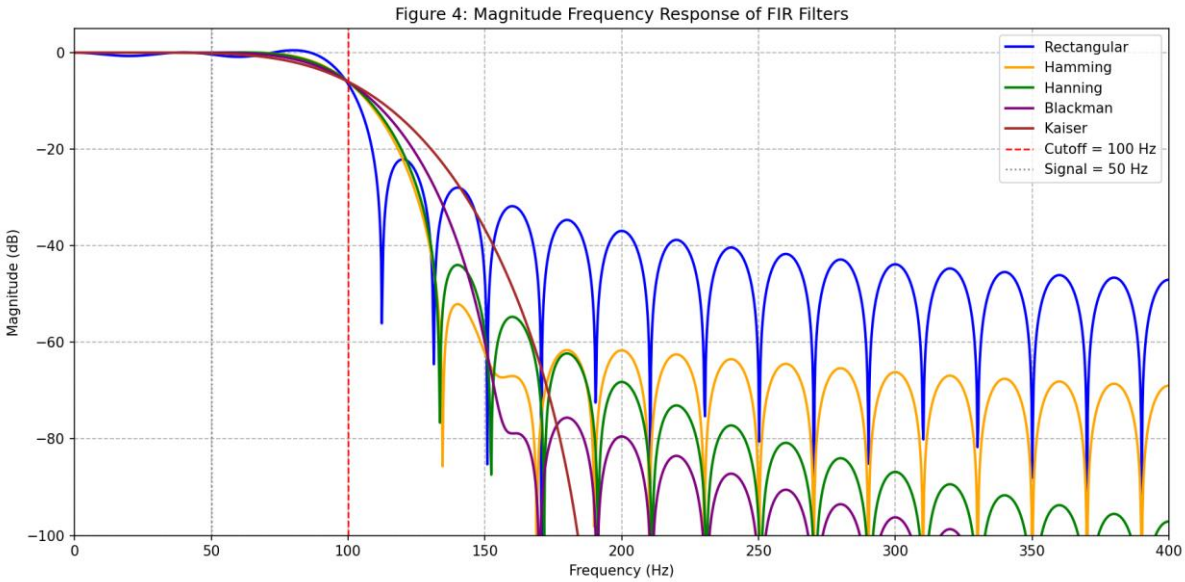


Figure 3 *shows the error signals for each window, calculated as the difference between the original clean signal and the filtered output. Lower error indicates better noise reduction. The Kaiser window shows the smallest error amplitude, while the Rectangular window shows the largest, confirming that Kaiser performs best and Rectangular least.*

Figure 4: Frequency Response of FIR Filters

Figure 4 shows the magnitude frequency response of each FIR filter. The Rectangular window exhibits the sharpest transition but the poorest stopband attenuation (only -13dB). The Kaiser window with β=14 achieves approximately -78 dB stopband attenuation, which explains why it removes noise most effectively. Blackman and Hamming windows fall in between.



7.1 Quantitative Results

The following metrics were calculated after compensating for the **25-sample group delay** to ensure accurate comparison:

Window	SNR Before (dB)	SNR After (dB)	SNR Improvement(dB)	RMSE
Rectangular	3.20	9.48	6.28	0.2375
Hamming	3.20	9.62	6.43	0.2335
Hanning	3.20	9.63	6.43	0.2334
Blackman	3.20	9.72	6.52	0.2310
Kaiser	3.20	9.81	6.61	0.2287

The results indicate that the Kaiser window achieves the highest SNR improvement of 6.61 dB and the lowest RMSE of 0.2287, confirming its superior performance. The Rectangular window, with only 6.28 dB improvement, is the least effective for noise reduction applications in this study. This difference is attributed to the Kaiser window's superior

stopband attenuation at β = 14, which more effectively suppresses high-frequency Gaussian noise.

8. Discussion

The experimental results demonstrate a clear relationship between window characteristics and filter performance. Windows with higher side lobe attenuation consistently produced better noise reduction:

Window	Side Lobe Attenuation	SNR Improvement
Rectangular	-13dB	6.28dB
Hamming	-41dB	6.43dB
Hanning	-31dB	6.43dB
Blackman	-58dB	6.52dB
Kaiser	~ -78dB	6.61dB

The Kaiser window's superior performance can be attributed to its tunable β parameter, as described by Harris [1] and Oppenheim & Schafer [3]. For this experiment, β = 14.0 was selected, providing approximately -78 dB stopband attenuation, which is significantly higher than the Blackman window's -58 dB.

However, improved performance comes with computational costs. The Blackman and Kaiser windows require more coefficients to achieve comparable transition bands, increasing processing time. In real-time applications, this trade-off must be carefully considered.

Additionally, the fixed filter order (51) across all windows may not be optimal for each window type. The Rectangular window could potentially achieve better results with a higher order, though this would increase computational requirements.

9. Conclusion

In this project, FIR filters using different window functions were designed and compared for noise reduction. Based on the results, the following conclusions are made:

1. Kaiser window (β=14) gave the best performance with 6.61 dB SNR improvement and lowest RMSE of 0.2287
2. Blackman window is second best with 6.52dB SNR improvement
3. Hamming and Hanning windows performed similarly with 6.43 dB improvement each
4. Rectangular window was the least effective with 6.28dB SNR improvement
5. Higher side lobe attenuation leads to better noise reduction

Performance from best to least: Kaiser > Blackman > Hanning ≈ Hamming > Rectangular.

10. Limitations

1. Only synthetic signals were used. Real-world signals might behave differently.
2. A fixed filter order (51) was used for all windows. Each window may have an optimal order that differs from this value.
3. The filters implemented were static, non-adaptive filters.
4. Only Gaussian white noise was considered. Other noise types may yield different results.

11. Future Scope

1. Test with real-world signals (ECG, audio, speech)
2. Implement adaptive filters for varying noise conditions
3. Try other window types like Bartlett and Chebyshev
4. Compare with IIR filters
5. Implement on hardware like Arduino or Raspberry Pi.

12. References

[1] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proceedings of the IEEE*, vol. 66, no. 1, pp. 51–83, Jan. 1978.

[2] SciPy Development Team, "SciPy: Open-source scientific computing tools for Python," [Online]. Available: https://scipy.org/. Accessed: Apr. 2, 2026.

[3] A. V. Oppenheim and R. W. Schafer, Discrete-Time Signal Processing, 3rd ed. Upper Saddle River, NJ: Pearson, 2010.